# Winning the Moment, Not Just the Draw: Redefining SHL Face-off Value through Context-Weighted Event Data

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**Abstract.** In ice hockey, player performance on face-offs is traditionally evaluated using aggregate Face-off Win Percentage (FO%), a binary metric that ignores spatial context and tactical outcomes. By treating all draws equally, FO% fails to capture the true value of a face-off, focusing on initial puck possession over game-winning impact. To address this, we introduce the Face-off Impact Score (FIS), a context-weighted metric that evaluates the Net Expected Goals generated within a strict 10-second post face-off window. Using event data from the Swedish Hockey League (SHL), FIS adjusts for score differential, manpower, period and face-off zone location. Our analysis reveals that offensive zone face-offs yield immediate shot attempts five times more often than center-ice draws. This disparity confirms a deployment bias in traditional metrics. By adjusting for these factors, FIS separates raw possession from tactical execution. Ultimately, our results prove that the teams's performance within the 10-second window is a superior indicator of impact than the face-off win itself.

**Keywords:** Face-offs · Context-weighted data · Ice hockey · Sports Analytics

## 1 Introduction

Ice hockey is characterized by continuous and fluid play. This makes face-offs the game's primary restarts and especially relevant for strategic evaluation. Player performance in the face-off circle is typically summarized using aggregate Face-off Win Percentage (FO%). This metric assigns equal weight to all draws regardless of context. It treats a neutral zone face-off early in the game the same as an offensive zone draw during a high-leverage power play. Specifically, this paper seeks to answer the following research question:

**To what extent does a context-weighted Face-off Impact Score (FIS) provide a more accurate evaluation of a player's face-off contribution than traditional binary win/loss percentages?**

## 2 Background

Traditionally, face-offs have been evaluated as isolated, binary events. However, modern analytics suggests that a face-off win is functionally meaningless if it is

immediately followed by a turnover. Its true value lies in the offensive potential of the resulting possession [1].

Silva et al. reinforce this by stating that possession alone does not guarantee scoring chances. This justifies evaluating face-offs through the subsequent 10-second window, prioritizing how effectively a team converts possession into a threat over the win itself [2].

Our model terminates post face-off xG chains at a strict 10-second threshold. Research demonstrates that shot generation rates regress to baseline averages approximately 10 seconds following a draw, irrespective of the starting zone [3].

## 3 Algorithm

### 3.1 Selection and Preparation of Relevant Data

Event data was processed using *pandas* and *NumPy*. Each face-off event was assigned a unique sequence identifier based on its index, along with its corresponding timestamp. These attributes allowed every event to be associated with its most recent preceding face-off. A filtering step was then applied to isolate post face-off sequences. Events were retained if they satisfied the following conditions: (i) a valid face-off sequence identifier was present, (ii) the event occurred within 10 seconds of the associated face-off and (iii) the event occurred within the same period. These criteria combined was used to filter the data into a new CSV file for more streamlined processing.

### 3.2 Processing the Data

Filtered data was processed using the statistical programming language R to categorize the results of each face-off. The processing pipeline followed a three step logic, pair identification, role assignment and outcome classification.

To analyze how two teams interact the data was grouped into unique face-off events. Ensuring that all actions within the same 10-second post face-off window were treated as a single sequence. Within each sequence xG were combined to capture the total offensive output generated during that interval.

The rink was divided into five zones, offensive zone, neutral zone (offensive), center circle, neutral zone (defensive) and defensive zone. Based on these coordinates, each team was assigned a specific role for that sequence, attacking team or defending team.

To process and measure the impact of a face-off each 10 second sequence was assigned to one of twelve outcomes based on the highest-leverage event recorded in that window. These outcomes were scoring events" which includes goals and shots by either team, Offensive Pressure this includes missed or blocked shots and territorial outcomes looks at where the puck ended up and which team had control in which zone. By assigning each face-off to one of these outcomes, weighting and calculation could be done to measure an impact score.

### 3.3 Defining the Face-off Impact Score (FIS)

To bridge the gap between binary outcomes and dynamic game states, we introduce the Face-off Impact Score (FIS). FIS measures the net xG generated within a 10-second window, adjusted by a situational leverage factor ($L$):

$$FIS = (xG_{team} - xG_{opponent}) \times (W_{score} \times W_{period} \times W_{manpower} \times W_{zone}) \quad (1)$$

Unlike traditional metrics, FIS rewards the quality of the resulting possession; it penalizes "won" draws that lead to immediate turnovers and rewards "lost" draws where the defensive unit successfully negates the opponent's advantage. The leverage factor ($L$) scales the net xG based on tactical urgency.
**Score Weight** ($W_{score} = 1/(1 + |score_differential|)$) applies higher leverage as the score gap narrows, prioritizing close-game situations.
**Period Weight** ($W_{period}$) increases from 0.6 in the first period, up to 1.5 in overtime to reflect the rising "cost of error".
**Manpower Weight** ($W_{manpower}$) being 1.0 when it's even strength, and 1.3 when it's power play and shorthanded.
**Zone Weight** ($W_{zone}$) is 1.3 for end-zones, 1.1 for neutral zones (Off/Def), and 1.0 for center, reflecting our empirical finding in Table 1 that end-zone face-offs generate higher offensive output.

To evaluate individual player performance over a larger sample, a player's overall FIS is calculated as their average yield per draw. This is achieved by summing the leverage-weighted net xG of every face-off taken by the player and dividing it by their total number of face-offs. This per-draw average ensures a fair comparison of true efficiency, preventing high-volume players from artificially inflating their impact score.

## 4 Findings

### 4.1 The Deployment Flaw, Why Face-off Location Dictates Impact

Before evaluating individual player impact, it is essential to establish the macro-level baseline probabilities of the 10-second post face-off window. The analysis of the data shows a clear link between ice location and event generation, a factor that traditional metrics fail to capture. For the attacking team, an offensive zone draw yields an immediate shot attempt (either on goal, missed, or blocked) 25.7% of the time based on Table 1. In contrast, a center ice face-off results in a shot attempt on less than 5% of the time. Immediate offensive threat is heavily dictated by starting location. This data underscores the fundamental flaw of relying on aggregate Face-off Percentage (FO%). By treating a center ice win and an offensive zone win as statistically equivalent, FO% ignores the massive disparity in their respective expected outcomes. This systemic bias validates the necessity of the Face-off Impact Score (FIS).

Across the entire SHL dataset, aggregate FO% creates a noisy leaderboard where raw win rates frequently fail to align with a player's actual goal differential.

In contrast, FIS successfully differentiates players across the league, aligning player rankings with true tactical yield. Figure 4 in the appendix visualizes this divergence across the full league. The four quadrants, divided by league medians, reveal that a high aggregate FO% does not reliably predict a high FIS. Several annotated outliers reinforce this. Player ID 376368 combines an above-median win rate with the highest FIS in the league, while Player ID 38871 posts both a low win rate and severe negative impact. Most notably, the upper-left quadrant contains players with below-average win rates who generate positive tactical leverage-players that traditional FO% would systematically undervalue. This divergence between raw possession and actual impact is then most clearly illustrated when analyzing specific player anomalies within the specific zones.

### 4.2 Context Under Pressure, Evaluating Player Leverage by Zone

Figure 1 in appendix demonstrates that the offensive zone is the primary driver of immediate post face-off scoring. Our analysis indicates that a superior face-off win rate does not inherently translate to high-leverage offensive outcomes. This is best exemplified by comparing outliers, Player ID 501734 achieved the highest aggregate win rate, yet Player ID 376368 produced the highest aggregate Face-off Impact Score (FIS). While the broader dataset shows a baseline correlation between FO% and FIS, however raw win percentages lose their predictive validity at the top percentiles, proving inadequate for accurately ranking the top 10 most impactful players in the SHL.

Figure 2 in appendix highlights critical tactical disparities within the neutral zone. When evaluating the FIS leaderboard, the highest- and lowest-ranked players share nearly identical raw win percentages. However, their true impact diverges drastically in goal differential. The top-ranked player drives positive leverage by generating an offensive goal every 210 face-offs, whereas the bottom-ranked player presents a severe defensive liability, conceding a goal every 136 face-offs.

Figure 3 in appendix illustrates the high-pressure nature of the defensive zone, a structurally disadvantageous position where teams are highly susceptible to conceding goals. Despite these pressures, Player ID 609414 emerges as a significant positive outlier. Operating under heavy defensive leverage, this player not only maintains a 60% win rate but remarkably drives offensive transition, generating a goal every 184 draws. In contrast, Player ID 38871 exemplifies severe defensive fragility, posting a mere 24% win rate and conceding a goal every 31 face-offs. Most notably, comparing ID 609414 (60% FO%) with ID 336459 (57% FO%) perfectly highlights the importance of our metric. Despite a marginal difference of just three percentage points in raw win rate, ID 609414 generates a vastly superior tactical impact, proving that the execution of the win matters far more than the aggregate percentage.

In order to better understand the leverage of face-off zone location, a calculation of the average FIS was made. The findings confirms that average FIS is significantly higher in the offensive zone (0.0056) compared to the other zones

which had an average FIS of ($<0.0005$). Analytically, this implies that approximately 178 offensive zone sequences are required to generate 1.0 net xG, whereas the other zones require over 2000.

### 4.3 Practical Applications

The Face-off Impact Score (FIS) offers actionable value by shifting focus from binary wins to 10-second tactical execution. Coaches can move beyond relying solely on aggregate FO% by integrating FIS alongside it, using high offensive FIS centers when trailing, or high defensive FIS centers to secure exits when leading. FIS exposes market inefficiencies, allowing teams to avoid overvaluing low-leverage possession and identify undervalued players who drive true net xG leverage. Because FIS evaluates the 10-second window, it grades the entire team. A team with high FO% but low FIS indicates broken post-draw set-plays rather than individual center failure.

## 5 Summary of Key Ideas

Based on the analysis of the 10-second post face-off window, this research establishes three foundational conclusions.

Traditional FO% is fundamentally flawed because it treats all face-offs as statistically equivalent. The data proves that an offensive zone draw is mathematically five times more likely to result in an immediate shot attempt than a center ice draw. Evaluating players without adjusting for this zone-based baseline creates a massive deployment bias.

Securing initial possession off the draw does not guarantee offensive leverage or defensive stability. As demonstrated across all three primary zones, players with identical or even high raw win rates can yield vastly different true tactical outcomes. Traditional FO% frequently rewards inconsequential puck control that fails to penetrate the opposing defense.

By weighting the outcome of a face-off against its zone-based baseline and evaluating the subsequent 10-second window, FIS successfully isolates the true tactical leverage. It accurately differentiates players based on their ability to convert a face-off win into a tangible, high-danger scoring opportunity (or defensive exit), proving to be a superior evaluative tool compared to aggregate win percentages.

Ultimately, while traditional face-off metrics merely indicate which player is most likely to win a face-off, FIS accurately measures which player actually drives game-winning leverage for their team.

## 6 Future Directions

Face-offs remain a relatively underexplored aspect of ice hockey analytics and several extensions of the present study could be pursued. For instance, the

fixed 10-second post face-off window could be refined by introducing adaptive or context-dependent thresholds based on the flow of play.

The proposed Face-off Impact Score could also be extended by incorporating additional contextual variables to improve its sensitivity to game state. For instance, the leverage could take into account the skill difference between the face-off players for both teams, or the amount of time spent on ice before the face-off. Finally, future research may explore alternative modeling approaches, including machine learning techniques, to better capture the complex situation between face-off outcomes and offensive performance.

## 7 Code Availability

Code: https://github.com/MervanPalmer/Sports-analytics

## References


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## Appendix

The table below summarizes the distribution of the 8,450 validated face-off sequences. The layout pairs symmetrical zones (offensive/defensive and neutral offensive/defensive) to illustrate the empirical basis for our leverage weighing ($W_{zone}$). Each zone/role combination sums to 100%, though rounding may cause slight deviations. Due to minor inconsistencies in the filtering process, a marginal discrepancy of three sequences (<0.04%) exists between the total attacking and defending roles. This does not impact the overall validity of the FIS distribution. In contrast to the offensive and defensive zones, the roles for center ice face-offs were determined by the winner of the draw.

**Table 1.** Macro-level distribution of 10-second post face-off outcomes, illustrating the zonal hierarchy across all primary zones.

| Zone & Role | Outcome | N | % | Zone & Role | Outcome | N | % |
|---|---|---|---|---|---|---|---|
| **1. Off. Zone** | 01. Goal (Own) | 64 | 1.06 | **5. Def. Zone** | 01. Goal (Own) | 12 | 0.20 |
| (Atk. Team) | 02. Shot on Goal | 742 | 12.33 | (Def. Team) | 02. Shot on Goal | 95 | 1.58 |
| | 03. Shot Attempt | 804 | 13.37 | | 03. Shot Attempt | 43 | 0.72 |
| | 04. Poss. Off. Zone | 1202 | 20.00 | | 04. Poss. Off. Zone | 380 | 6.32 |
| | 05. No Puck Off. Zone | 449 | 7.47 | | 05. No Puck Off. Zone | 1271 | 21.14 |
| | 06. Poss. Neut. Zone | 308 | 5.12 | | 06. Poss. Neut. Zone | 330 | 5.49 |
| | 07. No Puck Neut. Zone | 409 | 6.80 | | 07. No Puck Neut. Zone | 387 | 6.44 |
| | 08. Poss. Def. Zone | 838 | 13.94 | | 08. Poss. Def. Zone | 871 | 14.49 |
| | 09. Shot Against (M/B) | 42 | 0.70 | | 09. Shot Against (M/B) | 803 | 13.36 |
| | 10. No Puck Def. Zone | 1046 | 17.41 | | 09. No Puck Def. Zone | 1013 | 16.85 |
| | 11. Shot Against (SOG) | 95 | 1.58 | | 10. Shot Against (SOG) | 742 | 12.34 |
| | 12. Goal (Against) | 12 | 0.20 | | 11. Goal (Against) | 64 | 1.06 |
| **2. Neut. (Off)** | 01. Goal (Own) | 1 | 0.09 | **4. Neut. (Def)** | 01. Goal (Own) | 1 | 0.09 |
| (Atk. Team) | 02. Shot on Goal | 35 | 3.28 | (Def. Team) | 02. Shot on Goal | 17 | 1.59 |
| | 03. Shot Attempt | 19 | 1.78 | | 03. Shot Attempt | 12 | 1.12 |
| | 04. Poss. Off. Zone | 138 | 12.93 | | 04. Poss. Off. Zone | 105 | 9.84 |
| | 05. No Puck Off. Zone | 118 | 11.06 | | 05. No Puck Off. Zone | 151 | 14.15 |
| | 06. Poss. Neut. Zone | 81 | 7.59 | | 06. Poss. Neut. Zone | 95 | 8.90 |
| | 07. No Puck Neut. Zone | 113 | 10.59 | | 07. No Puck Neut. Zone | 99 | 9.28 |
| | 08. Poss. Def. Zone | 179 | 16.78 | | 08. Poss. Def. Zone | 292 | 27.37 |
| | 09. Shot Against (M/B) | 12 | 1.12 | | 09. Shot Against (M/B) | 19 | 1.78 |
| | 10. No Puck Def. Zone | 353 | 33.08 | | 09. No Puck Def. Zone | 240 | 22.49 |
| | 11. Shot Against (SOG) | 17 | 1.59 | | 10. Shot Against (SOG) | 35 | 3.28 |
| | 12. Goal (Against) | 1 | 0.09 | | 11. Goal (Against) | 1 | 0.09 |
| **3. Center Ice** | 01. Goal (Own) | 4 | 0.29 | **3. Center Ice** | 01. Goal (Own) | 3 | 0.22 |
| (Atk. Team) | 02. Shot on Goal | 35 | 2.56 | (Def. Team) | 02. Shot on Goal | 9 | 0.65 |
| | 03. Shot Attempt | 24 | 1.75 | | 03. Shot Attempt | 3 | 0.22 |
| | 04. Poss. Off. Zone | 222 | 16.20 | | 04. Poss. Off. Zone | 69 | 5.02 |
| | 05. No Puck Off. Zone | 93 | 6.79 | | 05. No Puck Off. Zone | 246 | 17.90 |
| | 06. Poss. Neut. Zone | 233 | 17.00 | | 06. Poss. Neut. Zone | 50 | 3.64 |
| | 07. No Puck Neut. Zone | 71 | 5.19 | | 07. No Puck Neut. Zone | 260 | 18.90 |
| | 08. Poss. Def. Zone | 245 | 17.90 | | 08. Poss. Def. Zone | 335 | 24.40 |
| | 09. Shot Against (M/B) | 3 | 0.22 | | 09. Shot Against (M/B) | 24 | 1.75 |
| | 10. No Puck Def. Zone | 427 | 31.20 | | 09. No Puck Def. Zone | 337 | 24.50 |
| | 11. Shot Against (SOG) | 9 | 0.66 | | 10. Shot Against (SOG) | 35 | 2.55 |
| | 12. Goal (Against) | 3 | 0.22 | | 11. Goal (Against) | 4 | 0.29 |

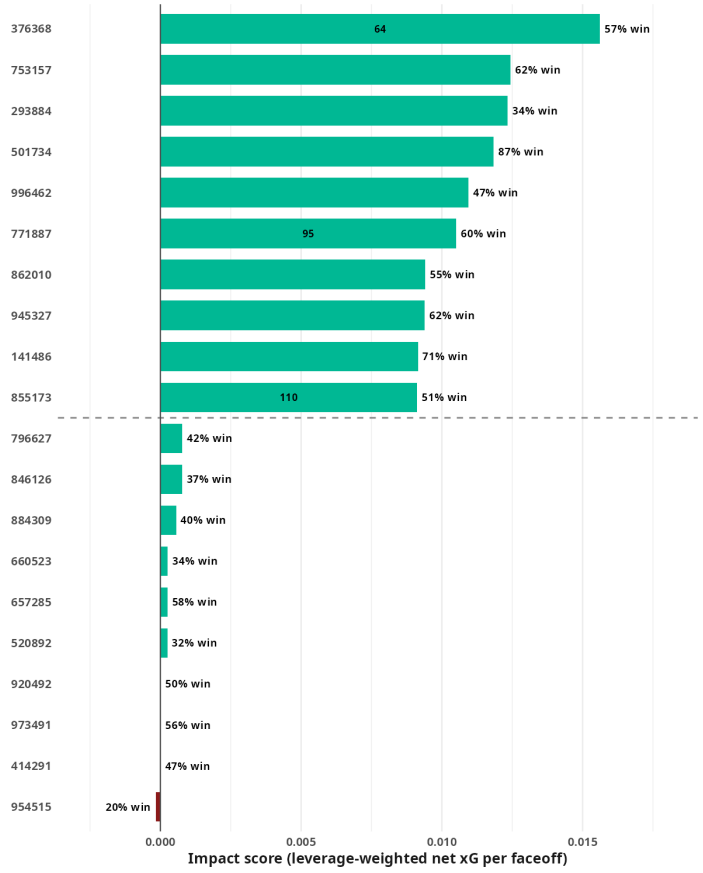


**Fig. 1.** Distribution of the top 10 highest- and lowest-impact players based on FIS in the **offensive** zone. The x-axis represents the impact score (leverage-weighted Net xG per face-off), while the y-axis represents individual players by their IDs.

The numerical label inside the bars for the highest-impact players indicates the average number of face-offs they must take to generate +1.0 Net Expected Goals (Net xG). In contrast for the lowest-impact players, it represents the average number of face-offs taken to concede -1.0 Net xG. The percentages displayed indicate the player's raw face-off win rate in this zone.

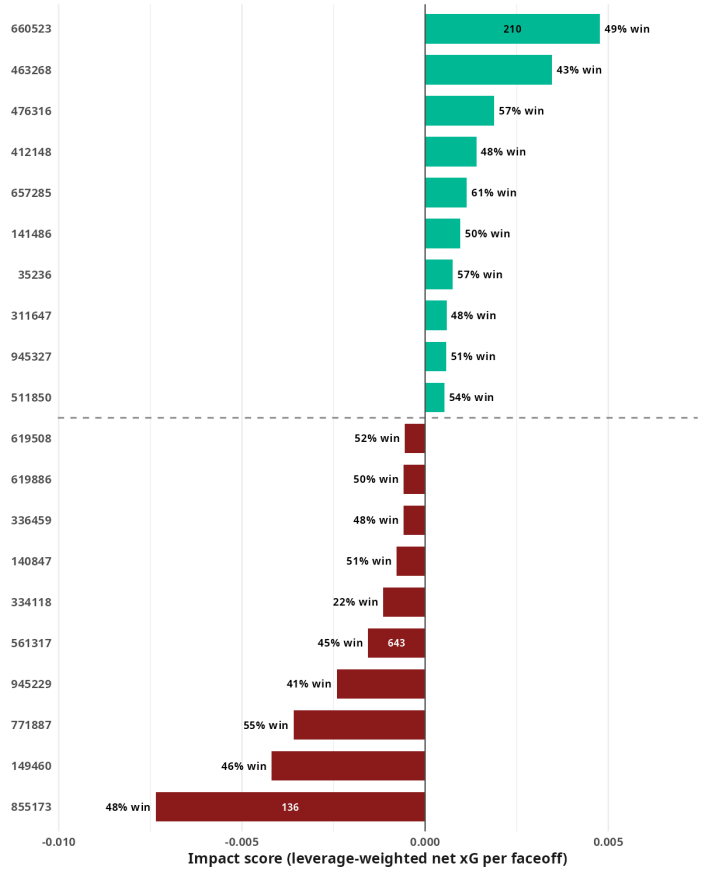


**Fig. 2.** Distribution of the top 10 highest- and lowest-impact players based on FIS in the **neutral** zone. The x-axis represents the impact score (leverage-weighted Net xG per face-off), while the y-axis represents individual players by their IDs.

The numerical label inside the bars for the highest-impact players indicates the average number of face-offs they must take to generate +1.0 Net Expected Goals (Net xG). In contrast for the lowest-impact players, it represents the average number of face-offs taken to concede -1.0 Net xG. The percentages displayed indicate the player's raw face-off win rate in this zone.

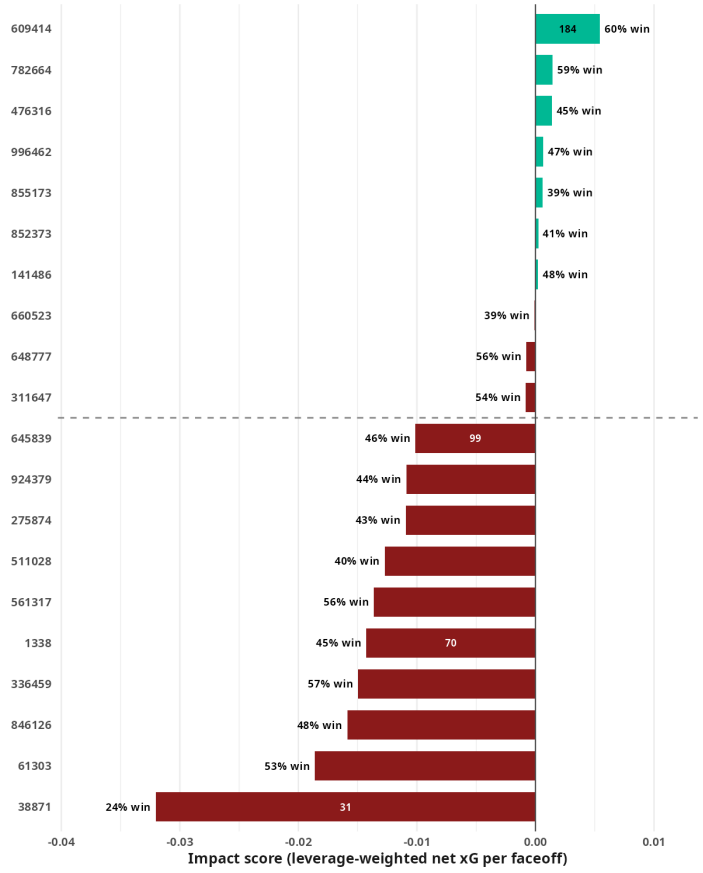


**Fig. 3.** Distribution of the top 10 highest- and lowest-impact players based on FIS in the **defensive** zone. The x-axis represents the impact score (leverage-weighted Net xG per face-off), while the y-axis represents individual players by their IDs.

The numerical label inside the bars for the highest-impact players indicates the average number of face-offs they must take to generate +1.0 Net Expected Goals (Net xG). In contrast for the lowest-impact players, it represents the average number of face-offs taken to concede -1.0 Net xG. The percentages displayed indicate the player's raw face-off win rate in this zone.

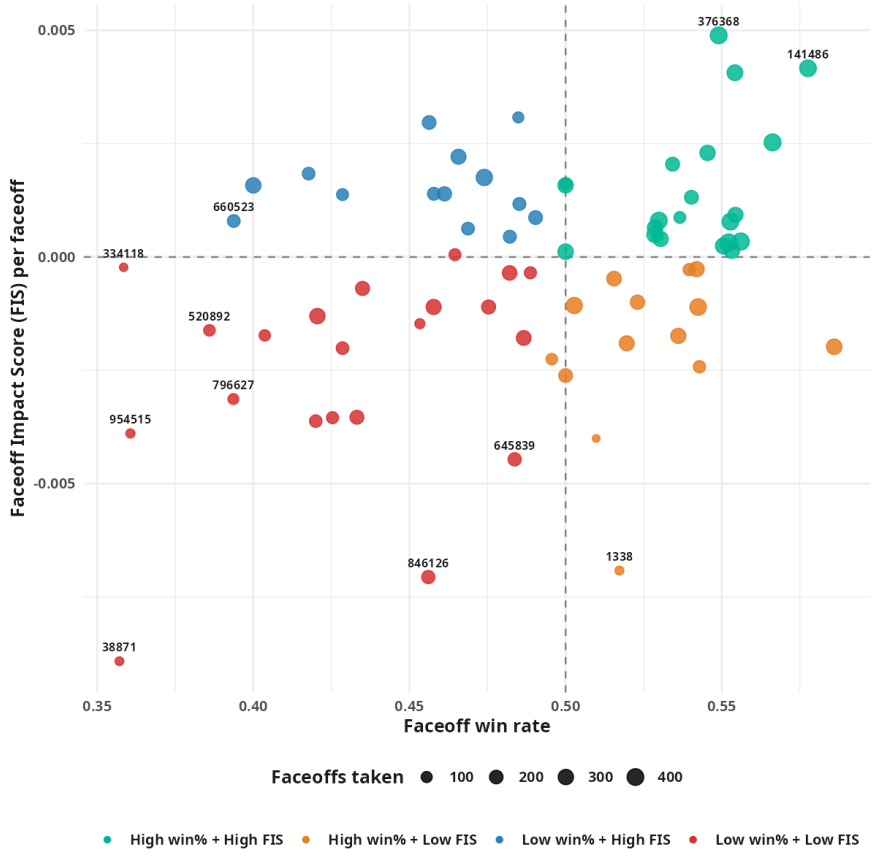


**Fig. 4.** Comparison of raw face-off win rate versus Face-off Impact Score (FIS) per draw. The x-axis represents the traditional aggregate win percentage, while the y-axis indicates the player's average tactical leverage (Net xG) generated per face-off. The size of each data point corresponds to the total volume of face-offs taken by the player.

The plot is divided into four quadrants based on league medians, visually demonstrating the divergence between raw possession metrics and true tactical output. Significant statistical outliers—players who deviate substantially in either aggregate win rate or FIS—are explicitly annotated with their unique player IDs.